DESIGN FOR SOCIAL INNOVATION

ASSIGNMENT - FROM CURRENT WORKFLOW TO FIELD WORKFLOW

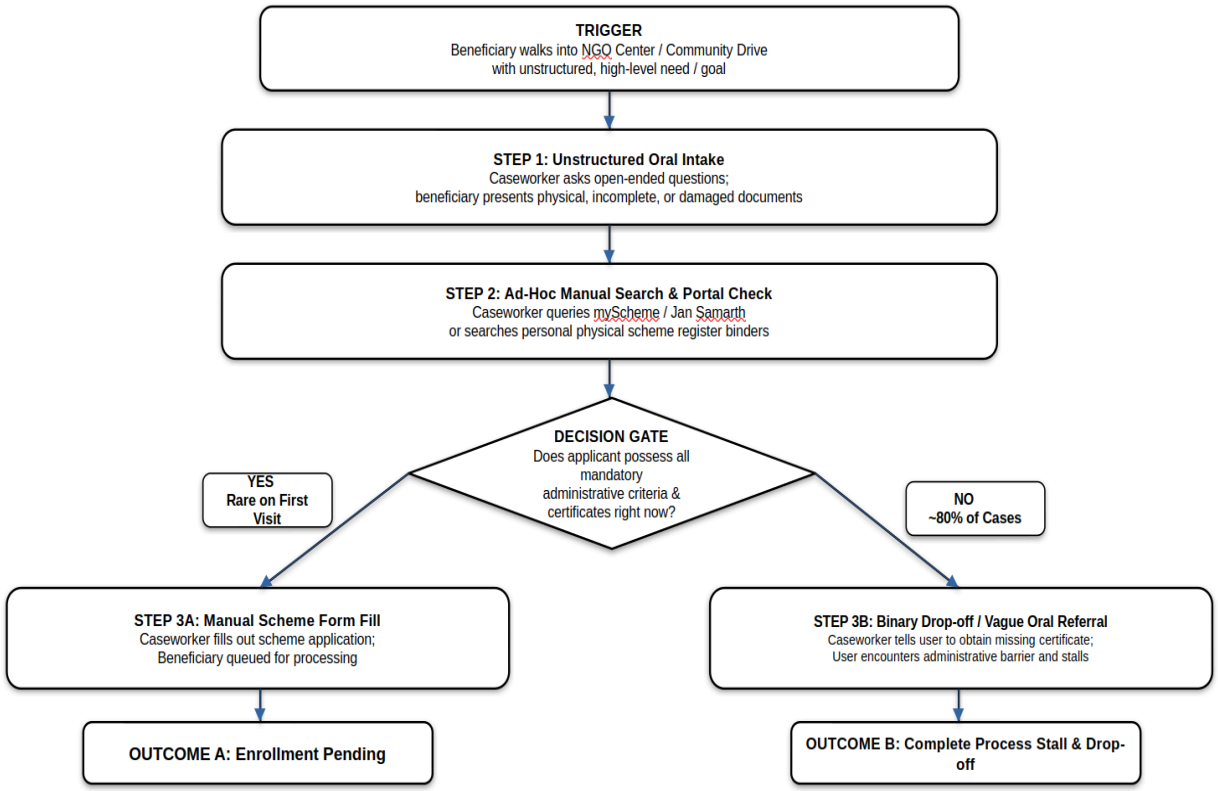
Section 1: Current User Workflow

1. Context & Operational Description

Today within partner NGOs (eg, Nirmaan skilling centers, Barabari Collective community drives), when a beneficiary arrives seeking support (eg, vocational skilling, scholarship or loan scheme), the process is entirely manual, narrative-heavy, and unscalable.

- **Trigger:** Beneficiary walks into an NGO center or mobile intake drive expressing a high-level aspiration (eg, "I want to enroll in the hardware technician course, but I have no income proof" or "I need a scholarship for my diploma").
- **Information & Inputs Used:** Unstructured verbal conversation; paper/physical identity cards (Aadhaar, old marksheet, ration card) often expired, undigitized or incomplete.
- **Caseworker Action & Decisions:** The caseworker conducts an ad-hoc, unstructured intake interview. To find eligible schemes, the caseworker relies on their own memory, handwritten binders, or attempts to navigate external portal catalogs like myScheme. These portals demand complex administrative parameters up front and issue a binary rejection if mandatory fields are blank.
- **The Failure Point:** The caseworker spends 35–50 minutes cross-referencing criteria across disconnected PDFs and websites. When a document is missing, the beneficiary is turned away with vague guidance without clear instructions on where to go, what supporting forms are needed or which local administrative kiosk (eg, MeeSeva/CSC) to visit.
- **Outcome:** High candidate drop-off, caseworker fatigue, slow triage throughput (only 5–8 cases per caseworker/day) and zero systematic tracking of whether the beneficiary ever completed the prerequisite action.

2. Current Workflow Diagram



What is Known from NGO Context: Caseworkers have high empathy but severe cognitive overload and beneficiaries possess un-digitized/missing records and lack administrative vocabulary.
Assumptions Requiring Validation: Caseworker intake time average (estimated at 40 minutes) and percentage of dropped cases directly caused by prerequisite confusion vs. logistical hurdles (transport/fees).

Section 2: Proposed User Workflow (Integrating the Product Core PoC)

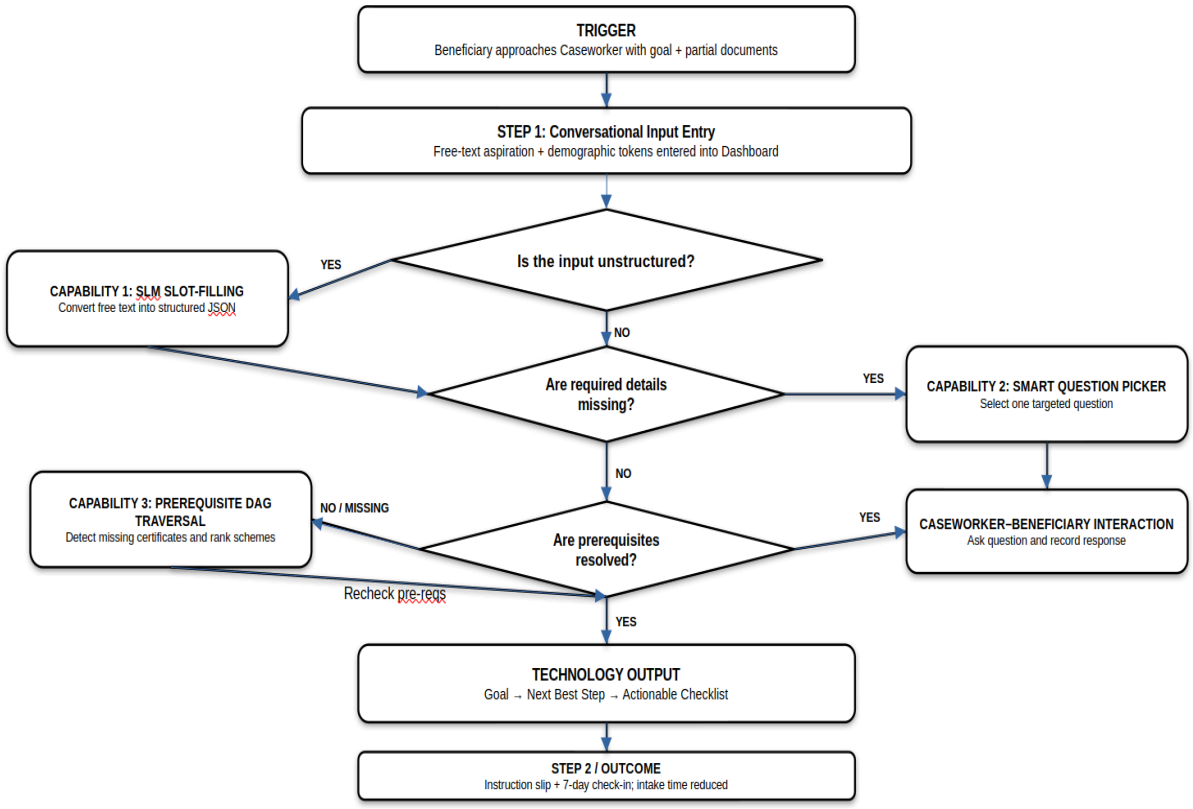
1. How the Solution Changes the Workflow

The proposed solution injects the Decision-Support and Active Information Gathering Engine directly into the intake interaction between the caseworker and the citizen. We do not replace the human caseworker; we provide them with an algorithmic triage tool that enforces structured prerequisite reasoning.

- **Where Technology Enters:** At the moment the beneficiary states their goal/need.
- **Input Provided:** Caseworker enters the beneficiary's free-text goal and whatever minimal attributes are immediately known into a low-bandwidth, mobile-first caseworker dashboard.
- **Core Technology Execution (The 3 Sprint 0 Capabilities):**
    - SLM Slot-Filling & Extraction: Quantized Small Language Model parses free text into structured JSON attributes.
    - Smart Question Picker (Entropy Minimization): Active questioning algorithm calculates information gain and prompts the caseworker with the single most impactful question to resolve candidate schemes, eliminating long forms.

    - Prerequisite Chain Traversal (Knowledge Graph DAG): NetworkX traversal identifies unmet dependencies (e.g., missing caste certificate) and finds the nearest achievable milestone rather than rejecting the applicant.
- **Technology Output:** A structured Visual Pathway Stepper displaying:
    - Confidence-Tiered Buckets: Confirmed, Likely (Pending Verification), Ruled Out.
    - The Immediate Next Best Step: An actionable instruction card (e.g., "Visit MeeSeva center with Ration Card + 2 photos to obtain Form 43 Income Certificate").
- **User Action After Output:** Caseworker prints or sends this localized instruction card directly to the beneficiary's phone via SMS/WhatsApp, scheduling a milestone verification check.

2. Proposed Workflow Diagram



Section 3: Potential Field Workflow

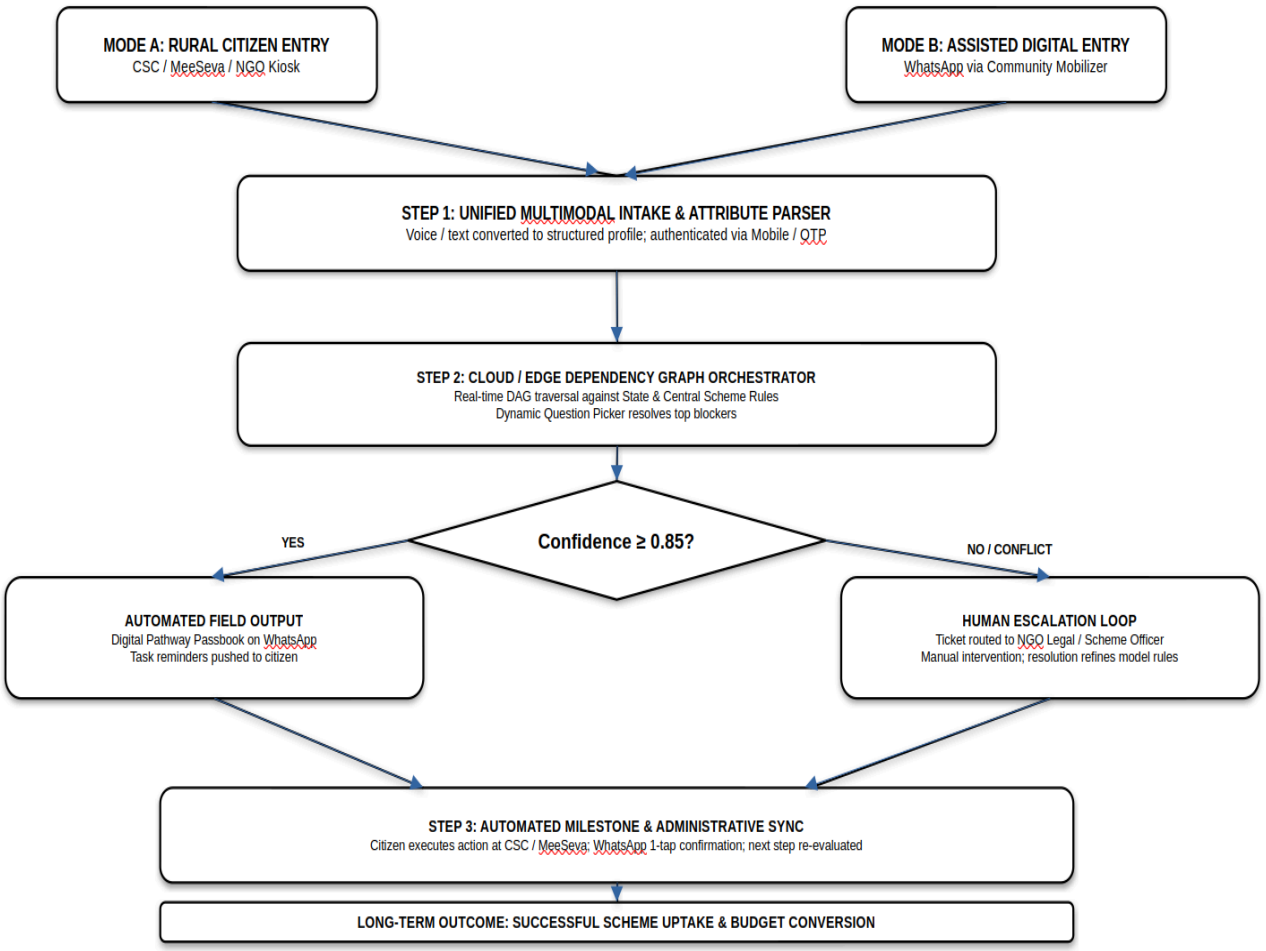
1. Looking Beyond the Semester PoC: Operational Field Deployment

In full field deployment across an NGO's statewide operations (e.g., 50+ Nirmaan livelihood centers and mobile rural vans), the core technology operates as a resilient, multi-stakeholder operational platform.

- **Actors & Modality:**
    - Primary Operator: Frontline Community Mobilizers and Village-Level Entrepreneurs (VLEs) at CSC/MeeSeva kiosks on budget Android tablets.

    - Direct Beneficiary Access: Self-service on WhatsApp/IVR for digitally literate citizens.
- **Frequency:** Continuous operational triage (20–40 cases processed per kiosk/day).
- **Exception & Failure Handling (Human-in-the-Loop Escalation):**
    - When ambiguous edge cases occur (e.g., uncodified local district exceptions, conflicting caste sub-categories), the system flags low confidence score and triggers an Escalation Loop, routing the ticket to a Senior NGO Legal Advisor or District Program Manager.
- **Organizational Changes in the NGO:**
    - Redefines the role of caseworkers from "information repositories" to "relationship and accountability mentors."
    - Requires a dedicated Scheme Rule Maintenance Desk within the NGO to update graph nodes when government departments alter scheme guidelines or deadlines.

2. Potential Field Workflow Diagram



Section 4: Three-Workflow Comparison

1. Comparative Parameter Table

Parameter	1. Current Workflow	2. Proposed Workflow (PoC)	3. Potential Field Workflow
Primary User	NGO Caseworker / Counselor (manual triage)	NGO Caseworker using mobile triage web-app	Frontline Mobilizer / CSC Operator / Citizen directly (self-service)
Key Task	Sifting catalogs & guessing next step manually	Guided attribute elicitation & prerequisite resolution	Automated pathway tracking, guided verification & escalation handling
Main Input	Free verbal dialogue + fragmented paper records	Stated goal + single-tap answers to smart questions	Multimodal voice input (Indic speech), location, & existing digital IDs
Role of Technology	External static portal queries (mySchеme) or none	Active decision support: SLM slot-filling, DAG prerequisite traversal, & question picking	Full operational orchestrator: Multi-agent triage, Bhashini voice layer, automated nudges & escalation routing
Main Output	Verbal opinion or flat rejection; unverified guidance	Visual Pathway Stepper: Confidence buckets + 1 concrete next step	Vernacular Digital Action Card on WhatsApp/SMS with localized center mapping

User Action After Output	Leaves frustrated or travels blindly to get documents	Caseworker hands citizen an actionable, single-task roadmap	Beneficiary completes mapped task; updates status via 1-tap voice/text prompt
People / Systems Involved	Caseworker, Beneficiary, Paper registers, myScheme portal	Caseworker, Beneficiary, Decision Support Tool (Next.js/FastAPI)	Beneficiary, VLE/Mobilizer, NGO Admin, CSC Kiosk, State Database APIs
Important Constraint	Caseworker cognitive limits, memory errors, candidate drop-off	Synthetic dataset boundary, rule coverage limits, web-only modality	Device bandwidth, digital literacy, local bureaucratic discretion, offline fallback

2. The Most Important Workflow Change: The single most transformative change is the shift from binary disqualification to prerequisite pathway sequencing. Instead of treating missing documentation as an immediate dead-end where citizens drop off, the technology actively diagnoses the missing dependency and converts it into a concrete, achievable milestone, transforming the caseworker from a manual catalog searcher into an empowering navigator.

Section 5: Assumptions & Questions to Validate

1. **Codification Ceiling of Unstructured Scheme Rules:**
   Assumption: Government welfare and skilling guidelines from unstructured sources (like *myScheme* text) can be reliably transformed into deterministic Directed Acyclic Graph (DAG) rules without requiring thousands of hours of manual legal engineering.
   Validation Method*:* Measure parsing accuracy and rule-conversion fidelity across a curated sample of 25 central/state skilling and welfare schemes with an NGO partner.
2. **Tolerance for Single-Question Active Elicitation:**
   Assumption: Asking 3 to 5 targeted, high-information-gain questions sequentially will reduce drop-off compared to a static upfront form, without irritating the user or caseworker through perceived slowness[cite: 4].
   Validation Method: Conduct controlled A/B mock intake sessions comparing full-form intake against dynamic single-question card flows with student and caseworker volunteers.
3. **Sufficiency of a Single Next Action:**
   Assumption: Providing a citizen with *only* their single nearest prerequisite milestone (e.g., getting an income certificate first) provides greater motivation and cognitive

clarity than showing them the entire, overwhelming 8-step bureaucratic pipeline upfront.
Validation Method: Qualitative user testing with low-income beneficiaries through Nirmaan/Barabari field workers to evaluate task comprehension and self-reported anxiety levels.

4. **Synthetic-to-Field Benchmark Gap:**
Assumption: An algorithm trained and benchmarked on synthetic profiles with artificially dropped attributes can reliably handle the messy ambiguity, conflicting records, and colloquial dialect phrasing of real-world community applicants.
Validation Method: Benchmark the algorithmic question picker against 50 anonymized historical case logs provided by partner NGO caseworkers.